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VITH SEMESTER B.TECH: [CSE]

END SEMESTER EXAMINATION (MAY. - 2019)

CO-304ARTIFICIAL INTELLIGENCE

Time: 3:00 Hours Max. Marks: 50

Note:	Attempt any five questions.
	All diagrams and pseudocode should be neat and clean.
	Assume suitable missing data, if any.

Q1(a) Explain different project characteristics and analyze the problem characteristics for robot moving a desk from one room to another? [5]

(b) Suppose two friends live in different cities on a map, such as the Romania map as shown in fig.1. On every turn, we can simultaneously move each friend to a neighboring city on the map.The amount of time needed to move from city i to neighbor j is equal to the road distance d(i, j) between the cities, but on each turn the friend that arrives first must wait untilthe other one arrives (and calls the first on his/her cell phone) before the next turn can begin.We want the two friends to meet as quickly as possible. Take the heuristic functions as 2 *D(i, j). Write the pseudo code for the problem using best first search and apply that on the problem to decide among the heuristic function. Assume one friend is at Oradea and other is at Neamt. [5]

Q2(a) Consider the given graph (Refer fig. 2) that represents an AND-OR graph. The terminal nodes are labelled SOLVED indicated by doubled rectangle and have zero cost. The arcs (all of them labeled 1) represent the cost of transforming the problem. Values associated with nodes are heuristic estimates of solving that node. Simulate the exploration of graph by AO^* algorithm till it terminates. Show how the graph looks at the end of each cycle. Assume FUTILITY value of 45. Clearly mark the final solution (by double-lined arcs) in the final graph. for this question. [5]

(c) Define alpha beta cut-offs? Show how the algorithm Alpha Beta explores the given game tree, searching from left to right. Refer Figure 3 [5]

(a) Conceptual Dependency (b) Syntactic analysis in NLP, (c) Logics for Non-Monotonic Reasoning.

Average {(10, 0.2), (20, 0.5), (30, 0.6), (40, 0.8), (50, 0.6), (60, 0.2)}
Rich = {(40, 0.2), (50, 0.5), (60,0.8), (70,1), (80,1)}
Draw the graphs of all three fuzzy sets on truth value such as very false, fairly false, false, true, very true. Find the membership value of salaries 30k and 60k [5]

Q6. (a) Given the following statistics, what is the probability that a woman has cancer if she has a positive mammogram result?

I. One percent of women over 50 have breast cancer.

II. Ninety percent of women who have breast cancer test positive on mammograms.

III. Eight percent of women will have false positives. [2]

(b) Explain the Bayesian Network [2]

(c) Explain the Process of Learning by using any neural network model or Genetic Algorithm? [2]

Q7. Write Short note on any of the two: [10]

(a) Conceptual Dependency, (b) Syntactic analysis in NLP, (c) Logics for Non- Monotonic Reasoning

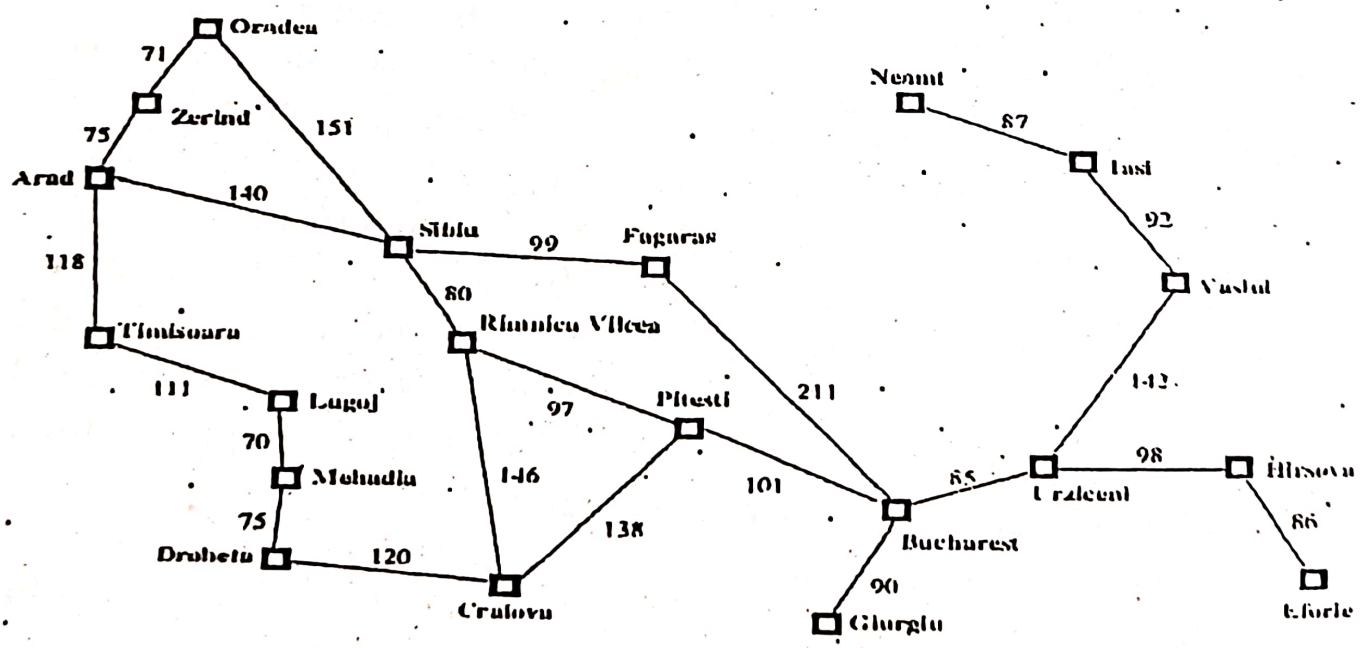

Figure 1

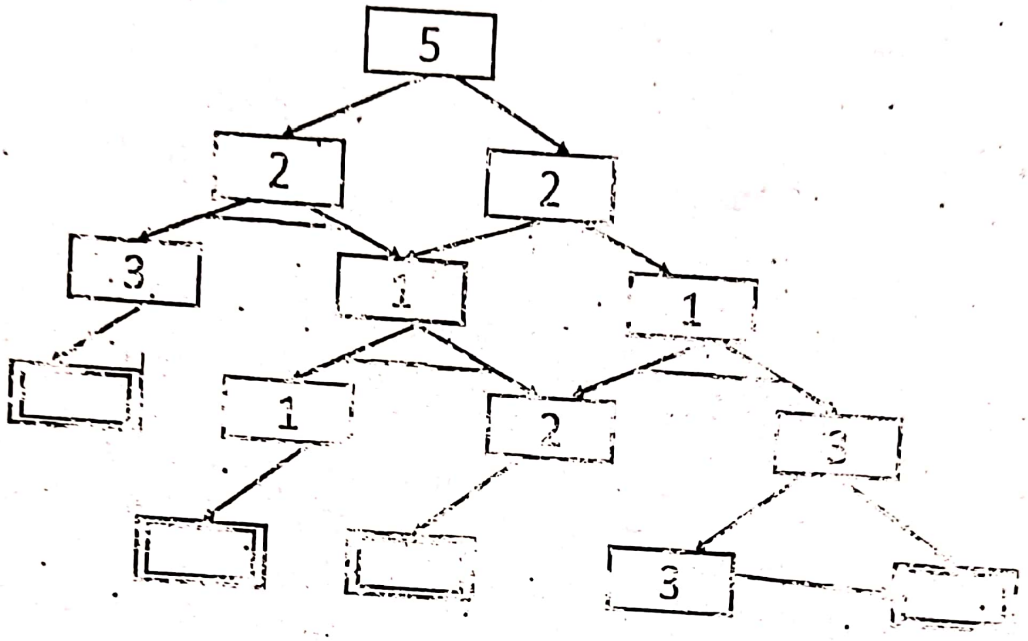

Figure2

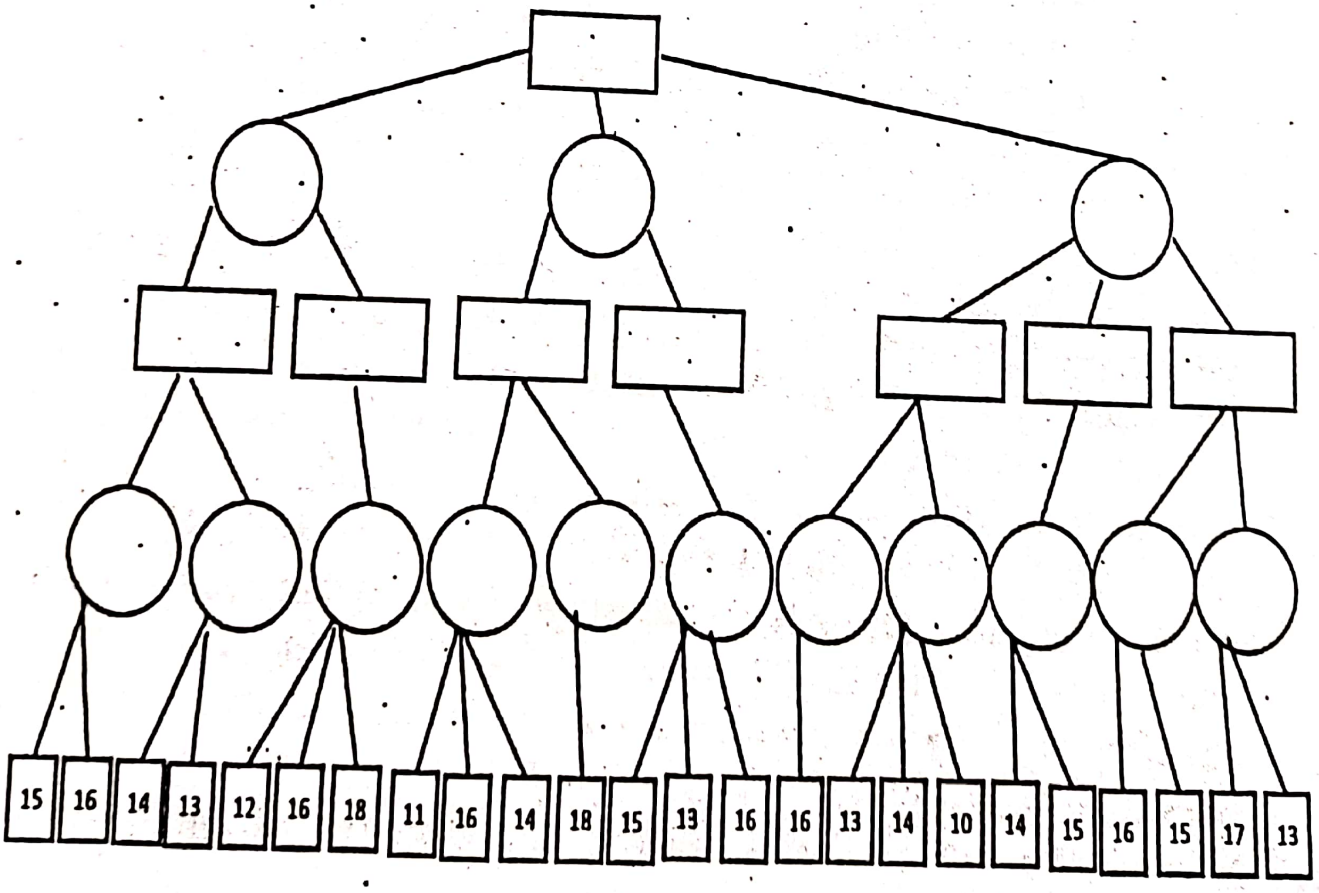

Figure 3

(i) Fill in the leaves that are inspected by AlphaBeta.Show the cutoffs and label them with their type.

(ii)Mark the move that AlphaBeta will choose for MAX at the root.

Q3. (a) Consider the following facts:

i. All students who are intelligent or hard working pass the exam.

ii. All students who pass the exam and have good communication skill get placement.

iii. Gaurav is intelligent and has good communication skill

iv. Gautam is neither intelligent nor hard working [10]

(b)

(i) Represent theses sentences as wff.

(ii) Using backward chaining prove that Gautam does not get placed.

(iii) Convert them to clause form indicating the law used.

(iv) Explaining steps of resolution, prove that gaaurav gets placed

Q4. (a) Every vehicle is a physical object consisting of engineand fuel system. Vehicles can be land vehicles having wheels, water vehicles having swimming system or air vehicles having flying wings. Land Vehicle can be road vehicle or railway vehicle. Air vehicle can be aircraft or space vehicle. Water vehicle can be river vehicle or sea vehicle. Road Vehicles can be cars having four wheels, or scooters having two wheels auto having three wheels. All cars have air condition and safety belt. Honda City is a car having 10 years of free service. Bajaj is a scooter having good average in terms of fuel. INS Vikrant is an instance of sea vehicle. MIG-21 is and instance of air craft. Dl-10c..3145 is an instance of car having red colour and model 2018.Represent the above knowledge using (i)Semantic Net and (ii) Semantic frame Write down the corresponding prolog representation and prolog rules to answer following queries.

a. Maruti has four wheels?

b. Vikrant cannot fly. [10]

Q5. (a) Describe the general architecture of expert system? How will you evaluate expert system like MYCIN? [5]

(b) Assume universe of discourse for the set of salary structures (in terms of K) per month is defined as U = {2, 5, 10, 20, 30, 40, 50, 60, 70, 80}. The Fuzzy sets defined on U based on salaries are:

Poor = {(2,1), (5,1), (10, 0.8), (20, 0.5), (30, 0.2)}

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